

Binayak Ojha

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Introduction

Passionate Computer Science Engineering graduate (B.Tech, 2021–2025) from C.V. Raman Global University with a strong academic record (CGPA: 9.0) and multiple research contributions in deep learning and cybersecurity.

Skills

- Python & Java
- AWS Cloud Technologies
- Data Structures and Algorithms
- Machine Learning & Deep Learning
- Data Science
- Networking & Business Analysis
- Advanced Java

Projects

- **Speech Emotion Recognition (Major Project):** Engineered a sophisticated deep learning model utilizing Convolutional Neural Networks (CNNs). Implemented advanced audio feature extraction techniques (MFCC, RMS, and ZCR) using Python and TensorFlow/Keras to accurately classify emotional states.
- **Company Management System:** Architected a scalable enterprise backend utilizing AWS (EC2, RDS) and advanced SQL. Focused on secure cloud integration, high availability, and robust relational data handling.
- **Disease Detection Using Machine Learning:** Built a predictive diagnostic system leveraging advanced machine learning algorithms (e.g., Ensemble methods, Scikit-learn) and Python data processing pipelines to accurately identify and classify medical conditions.
- **Online Taxi Management System:** Developed a reliable backend operational framework utilizing optimized RDBMS (PostgreSQL/MySQL). Focused on transaction management, secure data integrity, and efficient scheduling logic.

Certifications

- License Holder (Computer Engineer) at Nepal Engineering Council
- AWS Academy Graduate – Cloud Foundations
- CCNA: Introduction to Networks
- CCNA: Switching, Routing, and Wireless Essentials
- CCNA: Enterprise Networking, Security, and Automation
- DevOps on AWS Specialization

Internship

- Machine Learning Intern – Prodigy (1 month)
- Web Administrator Intern- Nice Pvt. Ltd. (2.5 month)

Education

B.Tech – Computer Science Engineering, C.V. Raman Global University, Bhubaneswar
2021–2025, CGPA: 9.0 (Completed)

Present Work

YSEF, Government Office under Ministry
Position:- IT Expert
(NOV 2025 - Present)

Research Publications

Conference Papers (Accepted and Presented)

- Deep Learning Fusion Ensemble for Enhanced Traffic Sign Detection Using the ICTS Dataset
- Enhancing Traffic Sign Detection Using a Hybrid Model of DenseNet121 and InceptionV3 with Optimized Dense Layers
- Advanced Traffic Sign Detection Using a Hybrid Deep Learning Architecture with Enhanced Dense Layer Customization
- Enhanced Emotional Speech Recognition with CNNs: Leveraging MFCC, RMS, and ZCR for Improved Accuracy (IISU-ASSET 2025)
- A Hybrid CNN-LSTM Approach for Sentiment Analysis on IMDB Movie Reviews

Journal Papers (Currently in Under Peer Review)

- Several Ongoing Researches
- Reviewer of various journals

Journal Papers (Published Online)

- Enhanced Framework for Credit Card Fraud Detection Using Robust Feature Selection and a Stacking Ensemble Model Approach (Result in Engineering)
- Analyzing the Effects of DDoS Attacks on Digital Twin-Enabled Industrial Systems: A Feature Selection with Particle Swarm Optimization based ELM Approach
 - <https://scholar.google.com/citations?user=cNFB8NcAAAAJ&hl=en>
 - www.linkedin.com/in/binayak-ojha-3477ba288